

**Inclusive Educational Software System: Emotionally Expressive
Reading for Children with
Disabilities (Blind, Deaf, Cognitive Impairments)**

Project ID: 25-26J-407

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in Information
Technology

Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

**Inclusive Educational Software System: Emotionally Expressive
Reading for Children with**

Disabilities (Blind, Deaf, Cognitive Impairments)

**Develop speech synthesis and Emotionally expressive text to speech and
user interaction**

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ABSTRACT

With the growing importance of digital environment, there is a critical need for educational tools that are inclusive and accessible to all children, including those with disabilities. Standard Text-to-Speech (TTS) systems, while useful, often lack emotional expressiveness, producing monotonous and robotic audio that can hinder comprehension and engagement, particularly for young learners. This project aims to address this gap by developing an emotionally expressive Text-to-Speech system with voice controls designed to enhance the story telling experience for children with visual impairments.

The primary goal is to create a dynamic, voice-engaging, and accessible TTS solution that can convey emotional tones such as happiness, sadness, and excitement, making educational content more relatable and easier to understand. The core of this research involves the fine-tuning of a voice synthesis model using emotionally annotated speech datasets. This model will be integrated into a user-friendly interface that provides interactive playback controls, allowing users to adjust audio parameters like speed, pitch, and volume to suit their individual needs.

This study will investigate the limitations of existing TTS technologies in educational settings and explore how emotional prosody can improve the story telling outcomes. By leveraging advancements in speech synthesis and machine learning, the proposed system will contribute to the development of more inclusive and effective assistive technologies, ultimately fostering a more companion environment for of the visually impaired children to gain an equal feelings for better emotional development.

Keywords: Text-to-Speech (TTS), Emotional Synthesis, Assistive Technology, Speech Synthesis, Accessibility, User Interaction, Inclusive Education

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DECLARATION

I hereby declare that the work presented in this report of proposal is my own and has not been submitted elsewhere. All sources of information have been duly acknowledged. This report is submitted in partial fulfillment of the requirements of the above degree program, and it reflects the research and development plan for my individual component of the project.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



Signature of the Supervisor

..... 31/8/2025.....

Date

Ms. Sanjeevi Chandrasiri



Signature of the Co-Supervisor

.....31/8/2025.....

Date

Ms. Karthiga Rajendran

LIST OF ABBREVIATIONS

Keyword	Meaning
TTS	Text-to-Speech
AI	Artificial Intelligence
ML	Machine Learning
UI	User Interface
NLP	Natural Language Processing

1. INTRODUCTION

The digital transformation of story teller has created immense opportunities for emotional development, yet it has also highlighted significant accessibility gaps. For children with disabilities, such as visual or cognitive impairments, standard digital resources can be challenging to navigate. Assistive technologies, particularly Text-to-Speech (TTS) systems, have become vital tools for converting written text into spoken words. However, the utility of these systems is often limited by their lack of emotional expression. The monotonous, robotic output of conventional TTS can make it difficult for children to stay engaged and fully comprehend the nuances of educational content.

Emotional cues in speech are fundamental to human communication, conveying context and meaning that text alone cannot. The absence of these cues in current TTS systems represents a major shortcoming. While research in speech synthesis has advanced significantly, the focus on creating emotionally dynamic and voice-accessible TTS for educational purposes remains an underexplored area. Mastery of technologies related to voice synthesis, machine learning models for emotion recognition in the voice, and user-centric interface design is essential for the successful implementation of this project.

The state of the art today includes several TTS engines with high levels of naturalness, but few are designed to dynamically adjust emotional tone based on the underlying text. Existing solutions rarely provide the interactive controls necessary for users with disabilities to tailor the experience to their needs. This research builds upon previous work by specifically focusing on integrating emotional expressiveness into a TTS system designed for an educational context. Unlike prior studies that focused primarily on the clarity of synthesized speech, this project zeroes in on the emotional delivery of content—a critical aspect for learning and engagement. By developing a system with emotionally aware TTS and voice accessible user controls, this project aims to create a tool that is not only functional but also deeply engaging and inclusive.

This research specifically addresses the distinct needs of its target users. For children with visual impairments, for instance, vocal emotional cues can act as a crucial interpretive layer, simplifying complex social information within a narrative and making the content more digestible and less overwhelming. Similarly, this system provides an emotionally expressive voice compensates for the absence of visual cues like illustrations or facial expressions, creating a more complete and immersive storytelling experience. This project therefore moves beyond basic functional accessibility to champion a form of

affective accessibility, where technology is not just a tool for information delivery but a medium for empathetic and context-aware communication. By focusing on this nuanced approach, the system aims to make story telling more intuitive and engaging, ensuring that the emotional depth of educational material is not lost in translation.

1.1 Background & Literature Survey

Accessible storytelling is a crucial aspect of inclusive digital culture, particularly in the context of modern media, where there is a growing emphasis on immersion, engagement, and user experience. The traditional digital storybooks, which often rely on static text or basic audio, are becoming increasingly inadequate in handling the diverse needs of young audiences, especially children with disabilities. Issues such as low engagement with monotonous narration, difficulty in understanding character emotions, and the potential for user fatigue highlight the limitations of conventional assistive reading technologies. As digital media consumption continues to accelerate, the need for more sophisticated, adaptive, and emotionally intelligent storyteller solutions is becoming more urgent.

The process of converting digital text into audible narration has traditionally been a functional operation, involving synthesized voices with neutral, robotic tones. This method is fraught with challenges, such as a failure to convey the author's narrative intent, a lack of immersion for young listeners, and an inability to make fictional worlds feel vibrant and alive. Furthermore, the reliance on basic Text-to-Speech (TTS) systems can introduce a significant barrier to enjoyment, making it difficult for children who rely on audio to connect with characters and plot on an emotional level.

The emergence of affective computing and advanced neural network-based speech synthesis offers a promising solution to these challenges by providing models that can generate narration with genuine emotional expression. By fine-tuning TTS models with emotionally annotated datasets, it is possible to create voices that convey tones like happiness, sadness, and excitement, eliminating many of the engagement and immersion issues associated with traditional systems.

The literature on assistive technologies has explored various approaches to enhancing digital media accessibility, each contributing to the understanding and development of more effective tools. However, there remains a significant gap in creating fully integrated narrator systems with a focus on emotional expressiveness, interactive user controls, and applicability in real-world storytelling environments for children with disabilities.

One of the research projects developed a highly natural-sounding TTS system using a neural vocoder, which focused on achieving near-human levels of clarity and intelligibility. This study highlighted the importance of high-fidelity audio in assistive technology to reduce listening effort. By using advanced deep learning models, the study was able to produce exceptionally clear speech. However, while the study successfully achieved naturalness, it did not explore emotional

expressiveness, which could have further enhanced the user's connection to the narrative.

Another research initiative focused on the integration of basic TTS into a children's e-reading platform. This system allowed for real-time text-to-speech conversion within an interactive storybook. While this approach improved basic accessibility, it still relied on a standard, monotonous TTS engine, which was not engaging for prolonged use and lacked the expressive range needed for effective narration.

The concept of emotional speech synthesis was further explored in a study that developed a prototype capable of generating speech in a few basic emotional states. This system utilized a complex model to manipulate prosodic features of speech based on emotion labels in the text. The study demonstrated that such a system could successfully produce recognizable emotions. However, while this approach addressed emotional expression, it was computationally intensive, had high latency, and was not integrated into an accessible, user-friendly application suitable for children.

In a more user-focused study, researchers developed an accessible media player with enhanced voice-controls for users with visual impairments. This system demonstrated the importance of a user-centric design in assistive technology, showing that customizable interfaces could significantly improve usability. However, this study primarily focused on interface design rather than the quality of the audio content itself. The study provided valuable insights into accessible controls but left the core issue of monotonous TTS narration unexplored.

This research aims to build upon the existing knowledge by developing an emotionally expressive narrator system complete with interactive voice controls. Unlike previous studies that either focused on audio clarity or interface design in isolation, this research will leverage advanced speech synthesis to create a system that is not only accessible but also deeply immersive and emotionally resonant. By doing so, it addresses critical gaps in the current literature, particularly the need for a system that can seamlessly integrate dynamic emotional expression with an intuitive, user-controlled interface.

The proposed system will reduce user fatigue, improve narrative comprehension, and ensure that the storytelling experience is enjoyable and captivating. This not only enhances the effectiveness of digital storybooks for children with disabilities but also increases their enjoyment and motivation, as they can be sure that the tools they use are designed with their needs in mind. Additionally, the integration of emotionally aware technology provides a scalable solution that can be adapted to various narrative platforms, making it a versatile tool for inclusive digital media.

The importance of this research lies in its potential to transform the landscape of digital storytelling by providing a more engaging, empathetic, and effective alternative to traditional narration methods. For authors and developers, this research offers a strategic opportunity to invest in innovative narrative tools that promise long-term benefits in terms of user engagement, accessibility, and artistic expression. For society, the successful implementation of this system could lead to more equitable access to culture and a higher overall quality of digital entertainment for children everywhere.

By developing a TTS system with emotional expressiveness and vocally emotional recognition for storytelling, this research not only contributes to the scientific development of assistive technologies but also addresses practical challenges faced by young readers with disabilities worldwide. It ensures that digital narrative systems are prepared to meet their demands.

The importance of this research lies in its potential to transform emotional storyteller systems by providing a more secure, transparent, and efficient alternative to traditional methods. For fund providers, this research offers a strategic opportunity to invest in innovative urban infrastructure that promises long-term benefits in terms of cost savings, increased efficiency, and improved public trust. For society, the successful implementation of this system could lead to more efficient use of technology resources, reduced congestion, and a higher overall moral support for visual impaired childrens.

By integrating emotionally richer text-to-speech tool to the story telling process with speech engine and voice control, this research not only contributes to the scientific development of emotional state but also addresses practical challenges faced by children with disabilities worldwide. It ensures that emotional story tellers systems are prepared to meet the demands.

1.2 RESEARCH GAP

Identifying research gaps is crucial for advancing the field and ensuring that new studies address unmet needs. Research gaps represent areas where existing studies have not fully explored specific aspects, leaving room for further investigation. In this study, the focus is on developing an innovative storyteller system that leverages emotionally expressive Text-to-Speech (TTS) technology to enhance the narrative experience for children with disabilities, particularly those with visual and cognitive impairments. A review of the existing literature reveals several research gaps that the proposed system aims to address.

The first research area focuses on high-intelligibility TTS systems[9] that use neural networks to produce exceptionally clear, human-like speech. While these papers discuss the importance of clarity for accessibility, they lack a detailed exploration of emotional expressiveness. The focus is mainly on the technical aspects of audio generation, without delving into how emotional prosody affects user immersion and connection to characters. While this work is foundational for natural-sounding speech, it does not cover how this speech could be adapted for dynamic, character-driven narrative content. Therefore, further research is needed to develop a narration system that combines high clarity with a rich emotional range.

A second area of research discusses digital storybook applications that include basic, built-in TTS functionality[3]. Although these papers offer a robust framework for delivering content, they do not explore the integration of advanced, dynamic TTS or provide users with sufficient controls over the audio narration. The TTS feature is often a simple read-aloud add-on rather than a core part of the narrative performance.

A third area of research discusses experimental prototypes of emotional TTS systems[4]. These systems often exist only in laboratory settings and are designed to test the capabilities of a specific algorithm. They lack an exploration of how such technology could be integrated into a real-world, user-friendly storyteller application for children. The focus is on the technical proof-of-concept rather than on practical usability, accessibility, or user experience. While these studies cover the generation of emotional speech, they do not integrate it into an accessible interface or test it with the target user group. Additionally, they lack discussion on interactive narration systems that allow users to customize their experience.

A fourth area of research showcases accessible UI designs for media consumption[10]. The proposed systems allow users with disabilities to easily navigate content and control playback.

While these studies address critical aspects of accessibility and convenience, they do not delve into the nature of the audio content being played. Additionally, these papers lack a discussion on how the interface could interact with an intelligent narration engine to modify a character's tone or the story's mood. Therefore, there is a significant research gap in exploring how an accessible UI can be integrated with an emotionally dynamic TTS engine to create a truly adaptive and personalized storyteller tool.

In conclusion, this project aims to address these identified gaps by developing a comprehensive storyteller system that integrates emotionally expressive TTS technology with an accessible, user-centric interface. By filling these gaps, the proposed system will contribute to the advancement of assistive technology, offering a more immersive, engaging, and user-friendly experience.

Research Gap Feature	Research 1 [9]	Research 2 [3]	Research 3 [4]	Research 4 [10]	Proposed Research Solution
Emotionally Dynamic Narration System					
Accessible User Interaction Controls					
Integrated Digital Storybook Application					

Table 1: Research Gap for component

1.3 RESEARCH PROBLEM

The current landscape of digital storytelling applications is riddled with several critical issues that severely undermine their effectiveness and inclusivity for children with disabilities. One of the most pressing problems is the emotionally sterile and unengaging narration due to a reliance on standard Text-to-Speech (TTS) technology. In many applications, the narration is generated by a monotonous, robotic voice, a method that is highly susceptible to misrepresenting the story's intended emotional tone. These inaccuracies can lead to a disconnect between the user and the narrative, frustration among young listeners, and ultimately, a diminished storytelling experience. Moreover, the flat nature of this audio is inherently unengaging, particularly for long-form content, resulting in a significant negative impact on the overall user experience.

In addition to these engagement inefficiencies, the accessibility vulnerabilities present in existing systems pose a substantial barrier. Traditional digital storybooks often lack robust, user-centric controls for audio playback, making them difficult for children with varying needs to operate. These vulnerabilities can lead to a frustrating inability to customize the experience, such as adjusting the narration speed or pitch, exposing children to a one-size-fits-all model that fails to meet their specific accessibility requirements and personal preferences.

Another critical issue is the loss of narrative context and richness. In conventional storyteller apps, users who rely on audio have little to no access to the emotional cues that are vital for understanding character motivations, plot development, and thematic depth. This lack of emotional expression can create a sense of confusion and detachment among users, who may feel that they are receiving an incomplete or "flattened" version of the story. The absence of a rich and dynamic auditory experience further exacerbates this issue, making it difficult for users to become fully immersed in the narrative world. [12]

Lastly, the unnatural cadence of robotic speech is a persistent challenge in many TTS-based systems. There is often a significant cognitive load associated with processing monotonous audio for extended periods, leading to listening fatigue.

These challenges highlight the urgent need for a more dynamic, immersive, and emotionally expressive storyteller system. By integrating advanced speech synthesis, which offers enhanced emotional range and user control, these issues can be effectively addressed. Such a system will enhance user engagement and satisfaction by providing a seamless, captivating, and empathetic storytelling experience.

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to design and develop an emotionally expressive Text-to-Speech (TTS) system featuring voice interactive user controls. This system aims to enhance the immersion, engagement, and accessibility of digital stories, while also ensuring a captivating and empathetic narrative experience for children with visually impairments.

2.2 Specific Objectives

1. Develop a TTS Module with Variable Emotional Expressiveness

This objective focuses on creating a secure and robust TTS engine capable of generating narration with distinct emotional tones (e.g., happy, sad, excited). It is **Specific** in targeting the synthesis of emotional prosody. The success will be **Measurable** by ensuring the generated audio is consistently rated as emotionally accurate in user perception tests. The objective is **Achievable** by utilizing and fine-tuning existing speech synthesis frameworks, **Realistic** in its integration with a web application, and **Time-bound** with a clear deadline for the core engine's completion.

2. Fine-Tune a Voice Synthesis Model for Enhanced Emotional Accuracy

The goal is to train and fine-tune a voice synthesis model using emotionally annotated datasets to improve the naturalness and accuracy of its emotional delivery. This objective is **Specific** to designing the model's training methodology, **Measurable** by the model's performance on validation datasets and its Mean Opinion Score (MOS) for quality, **Achievable** through the use of established machine learning platforms, **Realistic** given the availability of public datasets, and **Time-bound** to align with project milestones.

3. Develop an Accessible and Interactive User Interface

The objective is to create a seamless in-app interface that allows users to control the narration in real-time with voice. This is **Specific** to integrating voice interactive controls for playback speed, pitch, and volume. It is **Measurable** by ensuring over 95% of users can operate the controls without errors during usability testing. The objective is **Achievable** by utilizing standard UI/UX design principles and voice detection for accessibility. It is **Realistic** as it builds on established web application technologies, and it is **Time-bound** to be completed in alignment with the web's development timeline.

4. Implement User Feedback

The objective is to enable feedback notifications for the children through this system, providing instant satisfaction of the story and the system . This is Specific to enhancing user experience and transparency in the system's improvement. It is Measurable by ensuring that feedback trigger immediate notifications and responses. The objective is Achievable by integrating alert notifications and system response generation within the system. It is *Realistic* as these features are common in modern web applications, and it is Time-bound to be completed during the web development cycle.

3. METHODOLOGY

This research aims to develop an emotionally expressive and voice-centric Text-to-Speech (TTS) storyteller system. The methodology involves a structured approach, integrating key pillars of speech synthesis, machine learning, and web application development to achieve the project's main objective and specific sub-objectives. Each task and sub-task within the methodology is designed to ensure the system operates effectively, engagingly, and accessibly.

3.1 Key Pillars and Application of Technologies

1. **Speech Synthesis Technology:** Speech synthesis serves as the foundational technology for this, ensuring that all narrative text is converted into high-quality, natural-sounding audio. The advanced nature of neural network-based TTS prevents robotic-sounding output, which is critical for maintaining user immersion and engagement. By leveraging the strengths of modern TTS, the system will provide users with a clear, pleasant, and expressive platform where stories can be experienced audibly.
2. **Machine Learning (ML):** Machine learning models are self-improving systems with behaviors learned directly from data. In this project, an ML model will be fine-tuned to generate narration with emotional expression based on training with annotated speech datasets. The use of machine learning eliminates the need for manual, rule-based programming of emotions, providing voice emotion recognition and ensuring that the emotional delivery is nuanced, natural, and consistent.
3. **Web Application Development:** The **web application** acts as the user interface for the system, allowing users to interact with the emotionally expressive storyteller. Through the **web interface**, users can select stories, listen to the narration, and utilize interactive voice-active playback controls. The application will be developed to ensure a seamless user experience, integrating accessibility features and providing real-time control over the narration's speed, pitch, and volume.

3.2 Proposed Approach and Task Breakdown

1. Development of the Emotionally Expressive TTS Engine

The first major task involves developing the core speech synthesis engine that can generate narration with emotional expression and vocally interactive. This engine will be the heart of the storyteller, responsible for converting plain text into rich, vocally engaging audio.

- **Sub-task 1: Voice Synthesis Model Training**

A neural network-based TTS model will be fine-tuned on emotionally annotated speech datasets. This training will teach the model to associate text with the correct prosodic features (pitch, tone, pace) for various emotions.

- **Sub-task 2: Emotional Accuracy Validation**

Testing will be conducted to ensure that the model generates emotionally appropriate audio. This includes qualitative user testing and quantitative analysis to verify the naturalness and accuracy of the emotional expressions.

2. Development of the Accessible Web Application Interface

The next key task involves designing and building the user-facing web application. This interface will be simple, intuitive, and fully accessible, allowing children to easily interact with the stories and vocal narration controls.

- **Sub-task 1: UI/UX Design for Accessibility**

The user interface will be designed following Web Content Accessibility Guidelines (WCAG) to ensure it is usable by children with varying abilities. This includes creating large, clear controls and a simple navigation flow.

- **Sub-task 2: Voice Interactive Control Implementation**

The web interface will include voice controls for adjusting narration speed, pitch, and volume. These features ensure users can customize the listening experience to their personal comfort and preference and emotion tracking based on the voice.

- **Sub-task 3: Testing and Validation**

Testing will be conducted to ensure that the smart contracts function correctly under various scenarios. This includes testing different emotional outputs and interactive

methods.

3. System Integration and User Acceptance Testing

This task focuses on integrating the TTS engine with the web application and conducting thorough testing with the target user group. The end-to-end system must be seamless, reliable, and effective in delivering an immersive experience.

- **Sub-task 1: Backend and Frontend Integration**

An API will be developed to connect the web interface (frontend) with the TTS engine (backend). This will allow the user's text selections and control inputs to be processed in real-time to generate the correct audio output.

- **Sub-task 2: User Acceptance Testing (UAT)**

Testing sessions will be held with children to gather feedback on the overall usability, engagement, and effectiveness of the emotional narration. This feedback is crucial for validating the success of the project and identifying areas for improvement.

4. Implement Feedback Mechanisms and Performance Analysis

Continuous feedback and performance analysis are crucial for refining the user experience and validating the model's effectiveness. This task ensures that user insights are systematically collected and used to improve the storyteller.

- **Sub-task 1: In-App Feedback System Development**

An intuitive feedback system will be developed within the web application, allowing users to rate the quality and emotional appropriateness of the narration. This provides direct, contextual data for model improvement.

- **Sub-task 2: Performance Logging and Data Analysis**

After each user session, performance metrics and user feedback will be logged anonymously. This data will be stored and analyzed, providing a comprehensive and verifiable record of the system's effectiveness to guide future iterations.

3.3 System Diagram

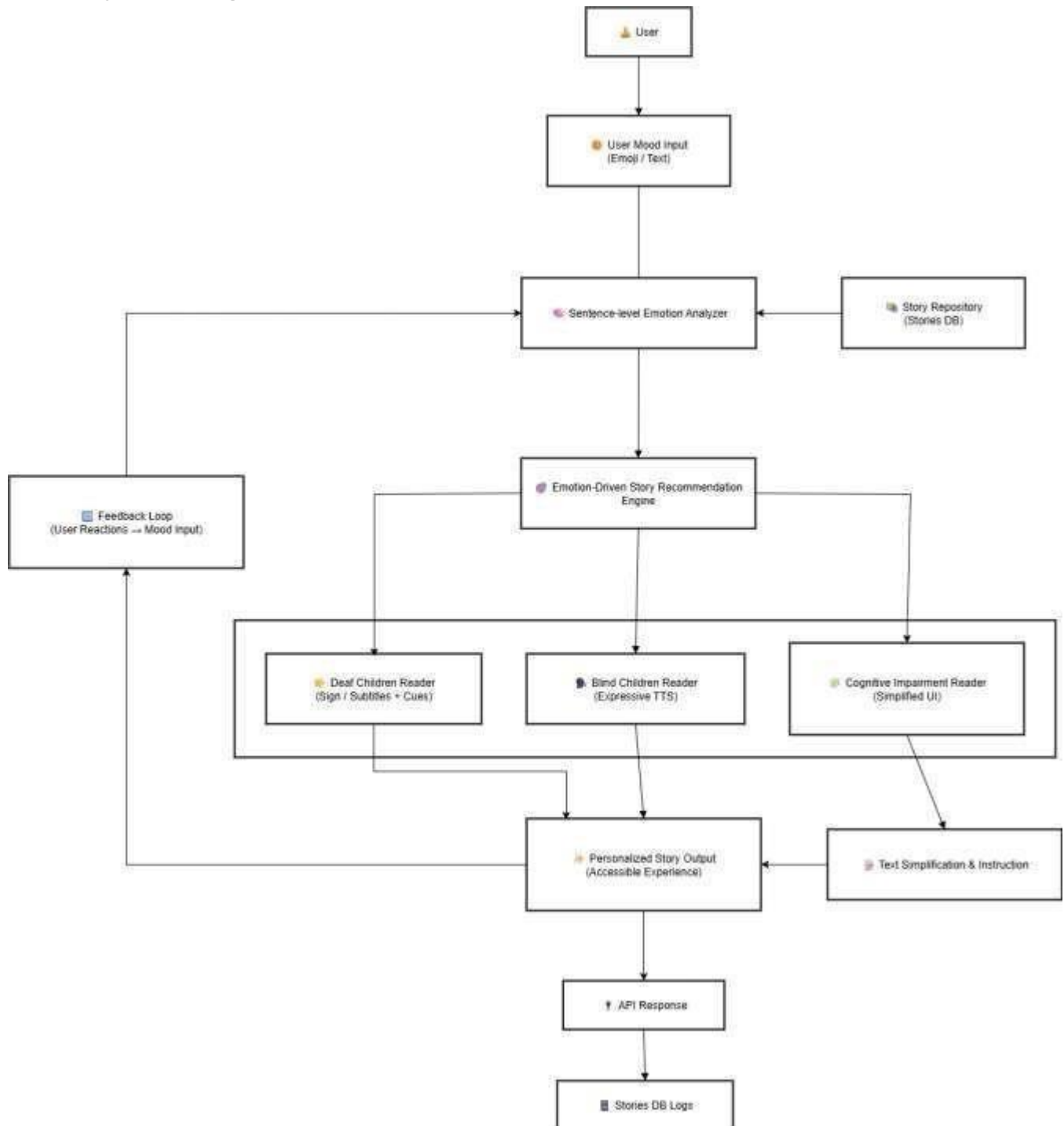


Figure 1: Component Diagram

3.4 Proposed Project Plan

3.4.1 Data Collection

The data required for this project includes emotionally annotated speech corpora, digital story texts, and user interaction feedback. The speech corpora will be used to train the TTS model. User feedback will be securely collected through the web application during evaluation. Machine learning ensures that the model's quality is auditable.

1. Speech Corpora (Training Data):

Data Needed: This includes high-quality audio files of speech labeled with corresponding emotional states, such as joy, sadness, or excitement.

Collection Method: Publicly available academic datasets (e.g., RAVDESS) will be sourced. These audio files and their annotations will be preprocessed and formatted to create a suitable training corpus for the synthesis model.

2. Story Content Data:

Data Needed: A variety of children's stories in digital text format, including narrative prose and dialogue, suitable for the target audience.

Collection Method: This data will be collected from public domain literary sources. The texts will be used as input for the TTS engine during the development, testing, and final demonstration phases of the project.

3. User Feedback Data:

Data Needed: Information regarding user ratings on emotional accuracy, narration quality, and the usability of the interactive voice active playback controls.

Collection Method: Feedback data will be collected anonymously from test users through the web application's integrated rating system. This data will be securely handled and analyzed to guide the iterative refinement of the system.

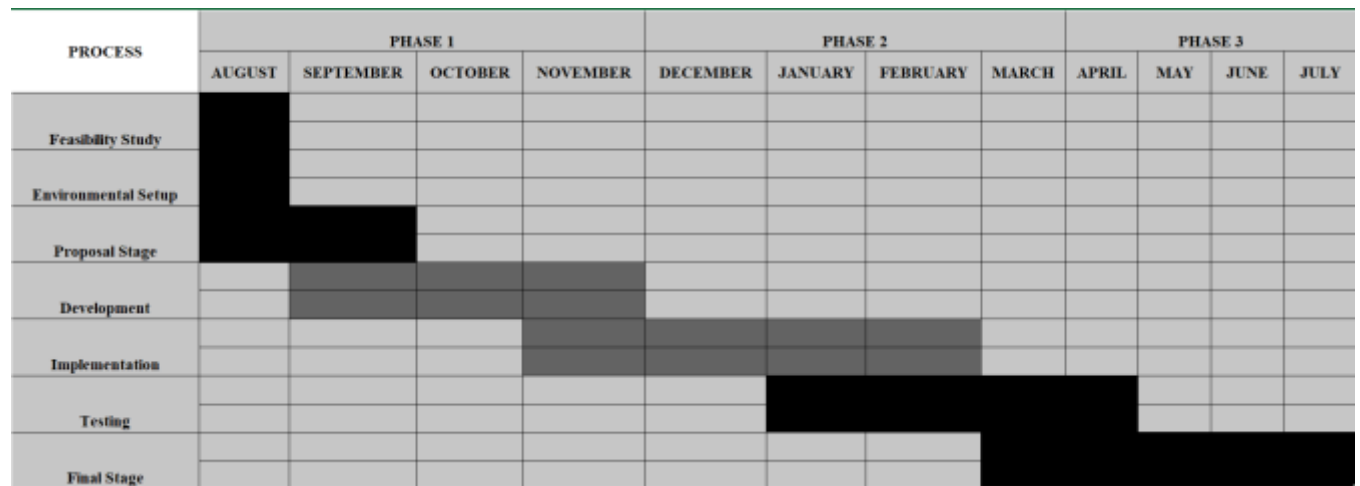


Figure 2: Gantt Chart

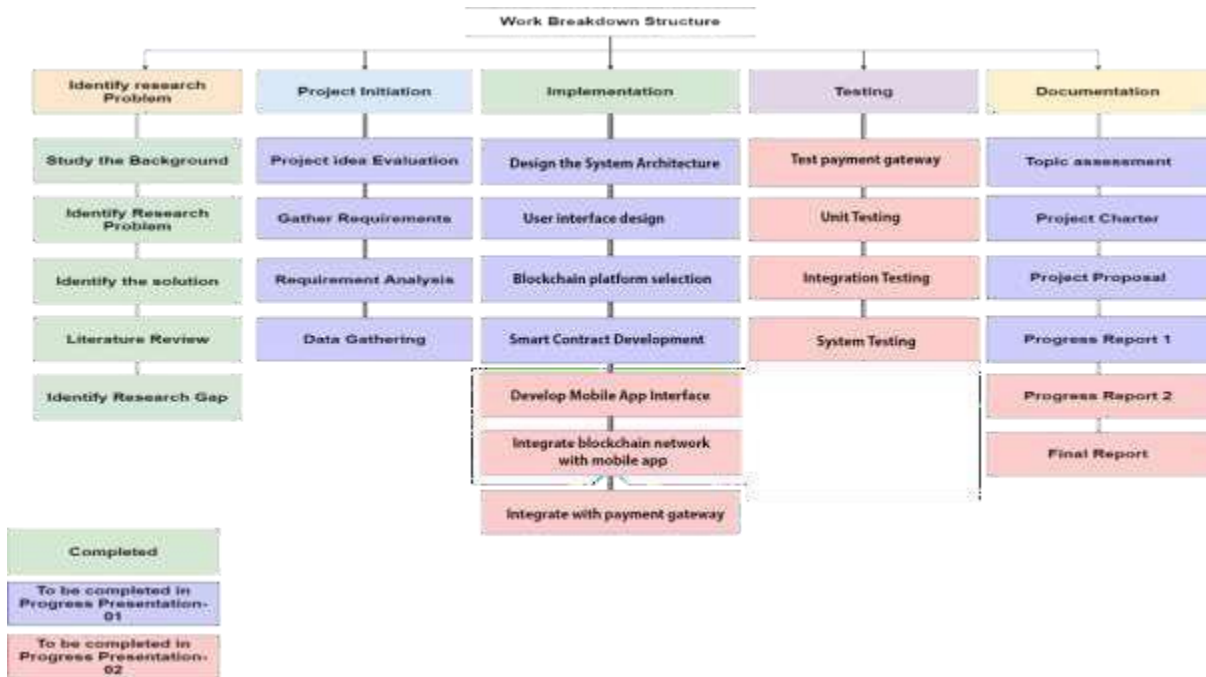


Figure 3: Work Breakdown Structure

3.5 Anticipated Conclusion

The anticipated conclusion of this project is the successful development and deployment of an engaging, immersive, and accessible emotionally expressive storyteller system. The system will significantly enhance the quality and reliability of automated narration, providing a robust solution to common issues such as listener fatigue and lack of emotional connection.

Results:

- **Accuracy and Expressiveness:** By leveraging a fine-tuned machine learning model, the system will ensure that all narration is delivered with appropriate emotional context, offering users a more authentic and transparent listening experience. The automation of emotional synthesis will minimize monotonous delivery, leading to more accurate and engaging storytelling.
- **Improved User Experience:** The web application will provide a user-friendly interface for selecting stories, controlling narration, and personalizing the experience. This will result in a more convenient and efficient storytelling platform for users, encouraging broader adoption among children with disabilities.
- **Enhanced Security:** The integration of real-time, user-centric controls for speed, pitch, and volume will protect users from a one-size-fits-all experience, reducing cognitive load and ensuring the narration is tailored to their individual needs.

Application:

- **Scalability:** The emotionally expressive TTS engine is scalable and can be expanded to serve multiple digital platforms. Its modular nature allows for easy integration with other inclusive software initiatives, enhancing overall digital content accessibility.

Real-World Use:

- **Inclusive Software System Integration:** The emotionally expressive narrator is specifically designed for implementation in the broader **Inclusive Educational Software System**, addressing the unique narrative needs of children with visual and cognitive impairments. The system will streamline the delivery of stories by automating expressive narration, reducing the comprehension gap, and providing rich, engaging audio content.

- **Enhanced User Engagement:** Children using the system will benefit from a user-friendly web application that allows them to easily select stories, listen to emotionally rich narration, and enjoy a more captivating experience by the voice controls. This will improve overall satisfaction and enjoyment, making digital storytelling more convenient and immersive.

In conclusion, this project will provide a cutting-edge solution for accessible digital media, addressing current challenges with innovative technology and paving the way for more empathetic and inclusive digital infrastructure.

3.6 Commercialization

The commercialization strategy for "**StoryVoice**" is designed to target key markets such as EdTech companies, digital publishers, special education institutions, and assistive technology developers. These sectors represent a significant demand for innovative narrative solutions that can enhance engagement, improve accessibility, and create a more immersive user experience. For these partners, StoryVoice presents considerable market advantages by making their content accessible to a wider audience, which in turn boosts user retention. The system's sophisticated narration engine ensures high-quality delivery, differentiating their products in a competitive market.

Economically, StoryVoice offers substantial value-addition for content providers by enabling them to serve a premium, inclusive market. The system's engaging narration can enhance user satisfaction and open new revenue streams. Its modular and scalable design allows for flexibility and future upgrades, making it easily adaptable as a licensed API for various digital platforms, from e-learning modules to interactive storybook apps.

By focusing on these core markets and emphasizing the dual benefits for both the end-users and content providers, the Emotionally Expressive Storyteller is well-positioned to become a leading solution in the market for advanced and accessible digital narration technology.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- **Text-to-Speech Conversion:**

The system must accurately convert digital text from selected stories into a clear and intelligible audio stream. This ensures that the core function of reading the text aloud is reliably performed for the user.

- **Emotional Expression Synthesis:**

The system must utilize the fine-tuned machine learning model to generate narration with varied emotional tones (e.g., happy, sad, excited) based on the story's text. The system must apply these emotions dynamically during narration.

- **Real-Time Narration Control:**

The web app must provide real-time, interactive controls for the user. Users must be able to see and adjust the narration speed, pitch, and volume during playback, providing full command over their listening experience.

- **Story Selection and Management:**

The system must present a library of available stories for the user to browse and select. Once a story is chosen, the system must load its text and prepare it for narration by the emotionally expressive TTS engine.

- **User Preference Persistence:**

After a user adjusts their narration settings, the system must provide a mechanism to save these preferences. The saved profile (e.g., preferred speed and pitch) must be automatically applied in future sessions, ensuring a consistent experience.

4.2 User Requirements

- **Ease of Use:**

The web application should be intuitive and easy to navigate, with a clean and responsive user interface that accommodates children of all technical levels and abilities.

- **Engagement and Enjoyment:**

Users expect the narration to be emotionally engaging and the emotion is recognized in the voice and the overall experience to be enjoyable, holding their attention throughout the story.

- **Accessibility:**

The web application must be accessible to users with disabilities, incorporating features like voice navigation, high-emotional conveying voice, and screen reader compatibility.

4.3 System Requirements

Hardware Requirements:

- Servers to host the web application and run the backend Text-to-Speech (TTS) engine.
- Client devices (like desktops, laptops, or tablets) with a modern web browser to run the web application.

Software Requirements:

- A backend framework for managing the TTS model and processing narration requests.
- Web application development frameworks for the user interface.
- A Machine Learning framework for the voice synthesis model.

Network Requirements:

- Reliable internet connection for real-time audio streaming from the server to the client's web browser.
- Sufficient bandwidth to ensure the audio stream functions well without buffering or delays.

4.4 Non-Functional Requirements

- **Performance:**

The system should process text and stream audio with minimal latency to ensure a smooth, uninterrupted listening experience.

- **Scalability:**

The system should be able to handle an increasing number of concurrent users and narration requests without performance degradation.

- **Security:**

The system must ensure any user data, such as saved preferences or feedback, is handled securely and stored privately to protect against tampering and unauthorized access.

- **Usability:**

The system's user interface, the web application, must be intuitive and easy for children to navigate, with simple controls and clear visuals.

- **Reliability:**

The system must provide consistent, high-quality audio narration, with a high availability rate to minimize downtime.

5. BUDGET AND BUDGET JUSTIFICATION

Component	Amount (LKR)
Web Hosting	9,600
Development Tools and Techniques	22,400
Cloud Hosting and Data Storage	12,000
Paper publishing cost	50,000
Total	94,000

Table 2: Budget Distribution

6. REFERENCES

- [1] M. E. Bausch and T. L. Sapp, *Understanding Assistive Technology: An Introduction to Students with Disabilities and their Educational Needs*. Pearson, 2008.
- [2] R. P. Silva, "Assistive Technology for Children with Learning Disabilities: A Systematic Literature Review," *ResearchGate*, 2023. [Online]. Available: https://www.researchgate.net/publication/368373667_Assistive_Technology_for_Children_with_Learning_Disabilities_A_Systematic_Literature_Review.
- [3] Y. Wang, R. Skerry-Ryan, D. Stanton, Y. Wu, R. J. Weiss, N. Jaitly, Z. Chen, Y. Zhang, Y. Wang, R. A. Saurous, D. Sculley, and R. C. Pang, "Tacotron: Towards End-to-End Speech Synthesis," in *Proc. Interspeech*, 2017, pp. 4006-4010.
- [4] J. Schröder, "Emotion in Speech," in *The Oxford Handbook of Affective Computing*, R. A. Calvo, S. D'Mello, J. Gratch, and A. Kappas, Eds. Oxford University Press, 2015, pp. 105-120.
- [5] E. Sri Ram, N. D. D. Deshmukh, and P. K. Ghosh, "Controlling Emotion in Expressive Speech Synthesis Using Latent Representations," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 7219-7223.
- [6] D. O'Hare and A. J. L. O'Hare, "The use of text-to-speech with children with reading disabilities: a systematic review," *Journal of Research in Reading*, vol. 44, no. 1, pp. 1-21, 2021.
- [7] F. Tits, K. El-Asmar, E. Van de Woestyne, B. Dutoit, and T. Dutoit, "Emotional Speech Synthesis for an Affective Storyteller," in *Proc. 8th ISCA Workshop on Speech Synthesis*, 2013.
- [8] J. Shen, R. Pang, R. J. Weiss, M. Schuster, N. Jaitly, Z. Yang, Z. Chen, Y. Zhang, Y. Wang, R. Skerry-Ryan, R. A. Saurous, Y. Agiomyrgiannakis, and Y. Wu, "Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram Predictions," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2018, pp. 4779-4783.
- [9] Y. Ren, Y. Ruan, X. Tan, T. Qin, S. Zhao, Z. Zhao, and T.Y. Liu, "FastSpeech 2: Fast and High-Quality End-to-End Text to Speech," *arXiv preprint arXiv:2006.04558*, 2020.
- [10] S. S. Lee, S. J. Cheon, and J. H. Kim, "Controllable Expressive Speech Synthesis Using a Domain-Adversarial Neural Network," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 1297-1307, 2020.
- [11] S. P. Macwan and A. P. Macwan, "A Review on Text-to-Speech (TTS) Synthesis Techniques for Visually Impaired Peoples," in *Proc. International Conference on Communication and Signal Processing (ICCSP)*, 2021, pp. 0177-0182.
- [12] P. N. Gonzalez, E. L. G. G. J. J. G. G. D. D. L. M. V. M. E. E. E. G. G. G. J. J. E. L. P. N. V. M. V. M. E. G. E. D. D. L. E. G. J. G. J. G. D. D. L. E. M. L. P. N. E., "Expressive Speech Synthesis with Tacotron2," in *Proc. IberSPEECH*, 2018, pp. 240-244.
- [13] P. N. Gonzalez, E. L. G. G. J. J. G. G. D. D. L. M. V. M. E. E. E. G. G. G. J. J. E. L. P. N. V. M. V. M. E. G. E. D. D. L. E. G. J. G. J. G. D. D. L. E. M. L. P. N. E., "Expressive

Speech Synthesis with Tacotron2," in *Proc. IberSPEECH*, 2018, pp. 240-244.

- [14] A. K. El-Assal, T. R. A. A. A. T. R., and H. E. S. D. D. D. A. H. E. S., "A Survey on Modern Text-to-Speech Synthesis Systems," *Ain Shams Engineering Journal*, vol. 13, no. 5, p. 101740, 2022.

